

When is a quantum system *hard* to simulate?

Based on [arXiv:2606.09971](#) and [arXiv:2609.XXXXXX](#)

PHYSICS COLLOQUIUM · NC STATE UNIVERSITY

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MOTIVATION

The four forces of nature!

We are governed by four fundamental interactions

Gravity stars & galaxies $\sim 10^{-38}$

E & M atoms, matter & light $\sim 10^{-2}$

Weak radioactive decay $\sim 10^{-6}$

Strong binds quarks & nuclei ~ 1

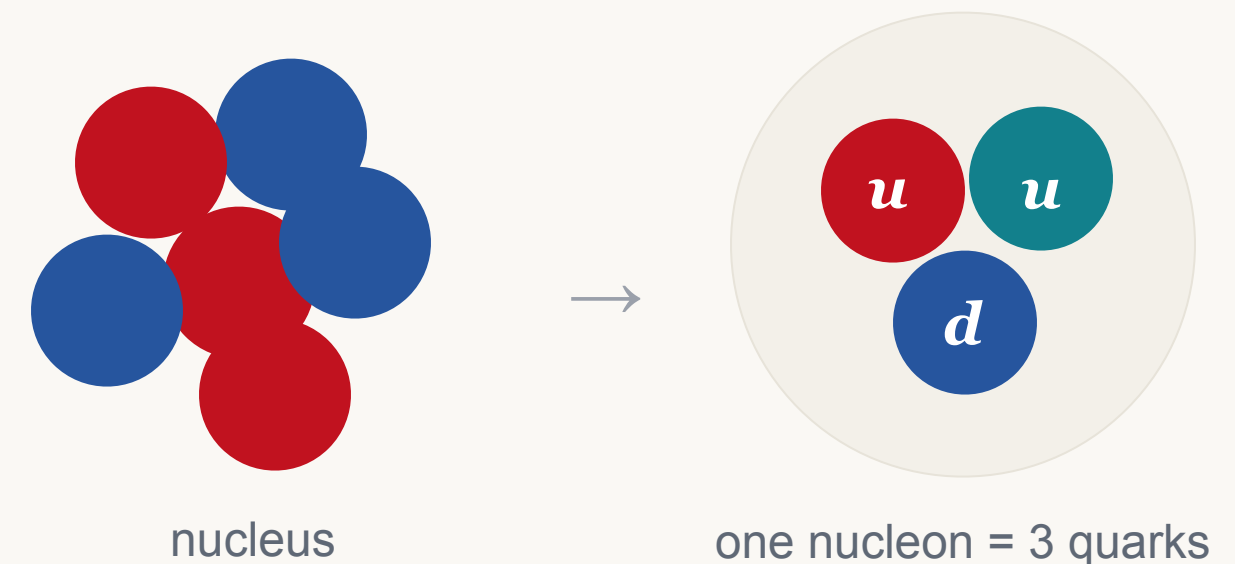
The **strong force** is so named because it is the strongest (no surprise!) — and the hardest to compute in the regime we want.

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Nuclear matter: the strong force at work

Protons and neutrons form the nucleus of every atom. Each nucleon is itself three **quarks**, glued together by the strong force.

Essentially **all** the visible mass in the universe lives here, inside nucleons.



The rules for these quarks and their glue are a theory called quantum **chromodynamics**

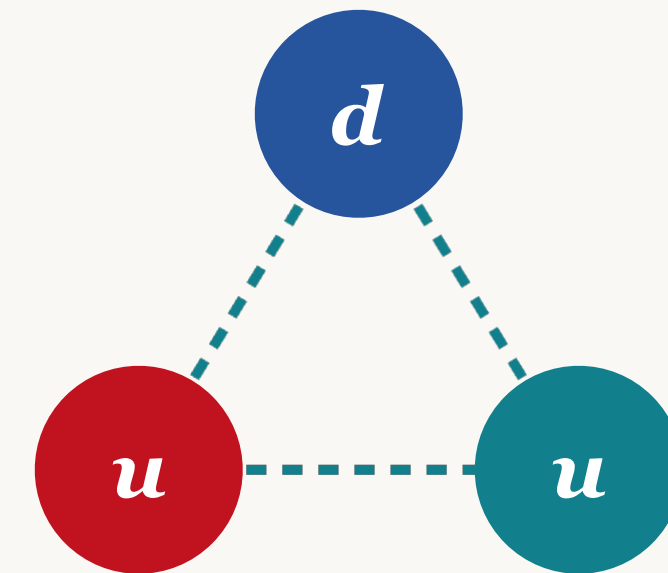
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Quantum Chromodynamics

The quantum theory of the strong force

- Quarks carry **color** charge (**red**, **green**, **blue**) and bind into **hadrons**.
- **Gluons** carry the force between them.
- **Confinement**: a single quark can never be isolated.
- **Asymptotic freedom**: at high energies, quarks behave as free.

At hadronic energies, QCD isn't solvable by hand — we have to go numerical.



proton = uud

Dalton thought atoms were indivisible; today quarks are the deepest layer of matter we know!

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Where is the proton's mass coming from?

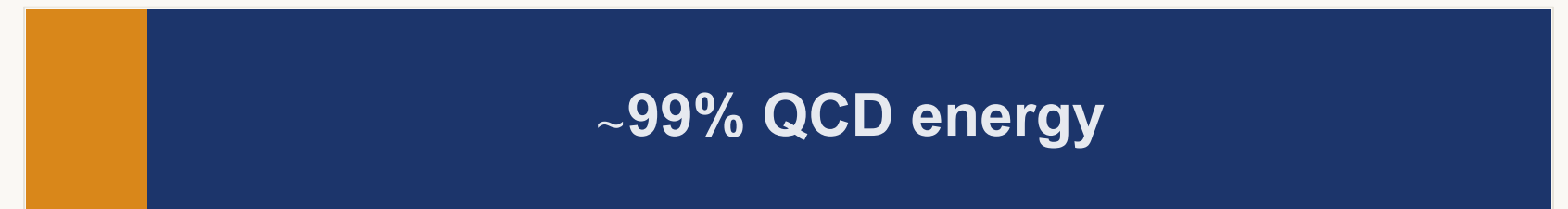
To understand this, understanding just quarks is not enough! We need to understand confinement of quarks, the interaction between quarks and gluons.

Wilson (1974) was the first one to realize that this can't be done with pen and paper!

He put QCD on a grid of spacetime points on a computer — birth of **lattice gauge theory**.

Simulated by Monte Carlo sampling methods in *imaginary* time, it reproduces the proton and other hadron masses **from first principles** to **~1–2%** (Dürr et al., *Science* 2008).

~1% Higgs



proton mass ≈ 938 MeV

up quark ≈ 2.2 MeV, down quark ≈ 4.7 MeV

$1\text{eV} = 1.6 \times 10^{-19}$ J

Only ~1% of the proton's mass is the quarks' own (Higgs) mass.

We want to understand the strong force. We can put it on a computer. It works. We can ask *any* question and then the supercomputer will tell us the answer! Why are we not done?



The catch: no access to Hamiltonian physics

Monte-Carlo based simulations of lattice QCD is based on action/Lagrangian — perfect for masses

But ask for **Hamiltonian picture**— scattering of gluons, how matter forms, dense nuclear matter dynamics— and it hits the notorious **sign problem**: the cost blows up **exponentially**.



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One (only?) alternative: quantum computers

“Nature isn't classical, ... if you want to make a simulation of nature, you'd better make it quantum mechanical.”
— Feynman, 1981

MAIN QUESTION

A quantum computer could do it in principle. But *how hard* is the job, really? Putting a number on that hardness is the central scientific question.



Optics News (1985).

Quick Review

Brief overview of classical and quantum approaches to computing

Classical computing: everything is 0s and 1s

A computer maps a string of **bits** — each a definite **0** or **1** — from one configuration to another.

- **Store:** information as a string of 0s and 1s.
- **Transform:** logic gates — AND, OR, NOT (**NAND** alone is universal).

A bit is *definitely* 0 or 1. N bits sit in exactly **one** of 2^N configurations at a time — a deterministic world.



a bit: one of two states



a string of bits

Quantum computing: a bit becomes a vector

A **qubit** is a vector in $\mathbb{C}^2$. The two classical values become basis vectors:

$$|0\rangle = (1,0)^T, |1\rangle = (0,1)^T$$

Unlike a bit, it can be in a **superposition**:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, |\alpha|^2 + |\beta|^2 = 1$$

- **Gates** are *unitary matrices* — always reversible.
- **Measure**: you get 0 with probability $|\alpha|^2$, 1 with $|\beta|^2$.
- Many qubits combine by **tensor product**:
 $|10\rangle = |1\rangle \otimes |0\rangle = (0,0,1,0)^T$

N qubits live in $\mathbb{C}^{2N}$ — the exponentially large space

Classical vs. quantum - at a glance

[Table from Jha, arXiv:2301.09679 quant-ph]

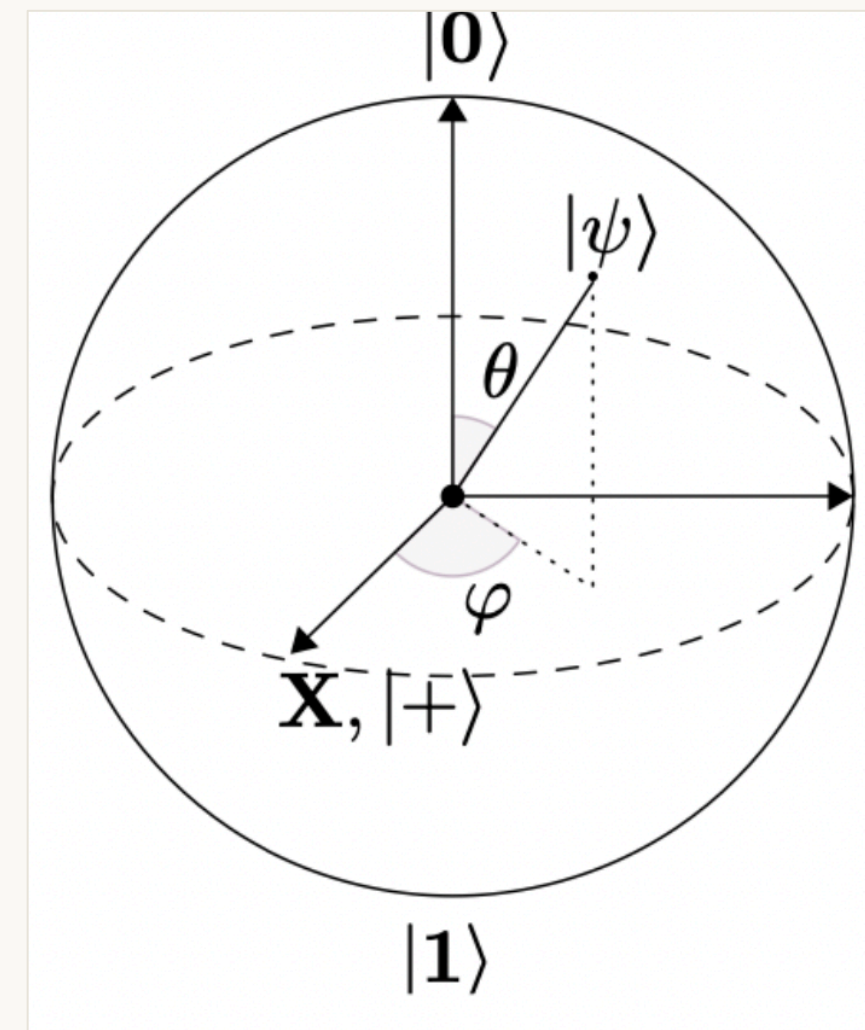
	CLASSICAL	QUANTUM
Algebra	Boolean	Linear
Basic unit	Bit	Qubit
States	Bit string	2^N -dim vector, $N \gg 1$
Gates	Logic gates	Unitary matrices
Reversible?	Sometimes	Always (unitary!)
Universal set	{NAND}	{ H , T , CNOT}

The universal quantum set needs the **T-gate (2x2 unitary matrix, $\text{diag.}(1, \exp(i\pi/4))$)** — the gate that supplies the **magic**.

The simplest quantum system as geometry

We just wrote a qubit as a *vector*, now let's look at the same thing as *geometry*. $|0\rangle$ and $|1\rangle$ are like north and south pole. Every other point on the surface is a bonafide state. Some states are harder to reach!

A **superposition** such as $(|0\rangle + |1\rangle)/\sqrt{2}$ then becomes a **point on a sphere** — the Bloch sphere — fixed by two angles θ, φ .



North pole $|0\rangle$, south pole $|1\rangle$, equator = superpositions.

Part I

“Quantum hardness”: the probe

What truly brings in the effectiveness of quantum computers?

Lesson from chemistry!

A diamond scratches glass; glass scratches a penny; your fingernail scratches chalk. The **Mohs scale** just ranks materials by how well they resist being scratched by the next one.

THE SCALE WE WANT!

Can we build a scale like this for **computation** — how much a quantum system resists being simulated by our classical computers?



1 · WHAT DO WE MEAN BY “HARD”?

Quantum systems live in astronomically big space

N classical bits need N numbers. N quantum bits need 2^N numbers — one amplitude per pattern.

N

numbers to describe N classical bits

2^N

amplitudes to describe N quantum bits

275 qubits

more amplitudes than atoms in the universe

“The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.”

— Dirac, 1929

1 · WHAT DO WE MEAN BY “HARD”?

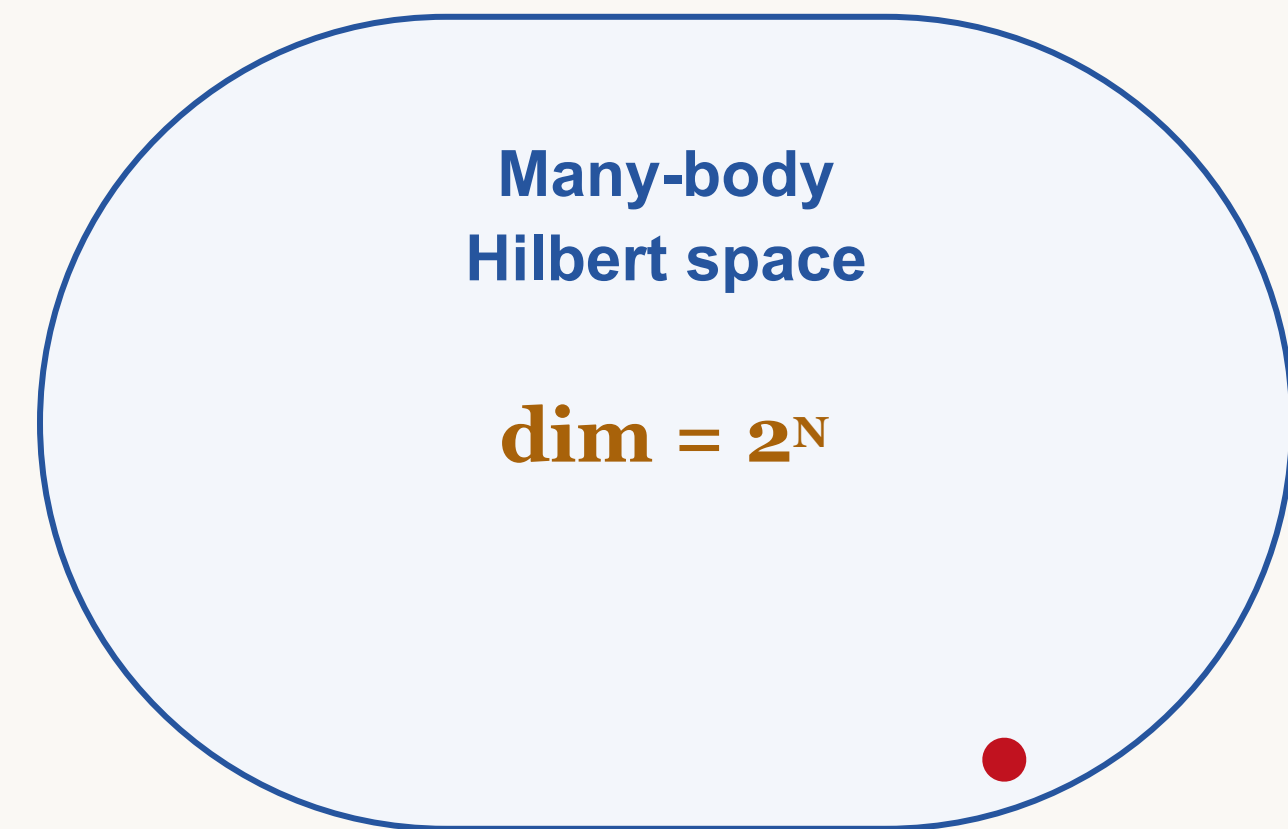
But nature is often kinder than exponential growth

The physics that actually matters sometimes lives in a **tiny corner** of that enormous space.

Classical methods like **tensor networks (TN)** are built to find and work inside that corner. When they can, a laptop beats a supercomputer!

“BIG” AND “HARD” ARE DIFFERENT

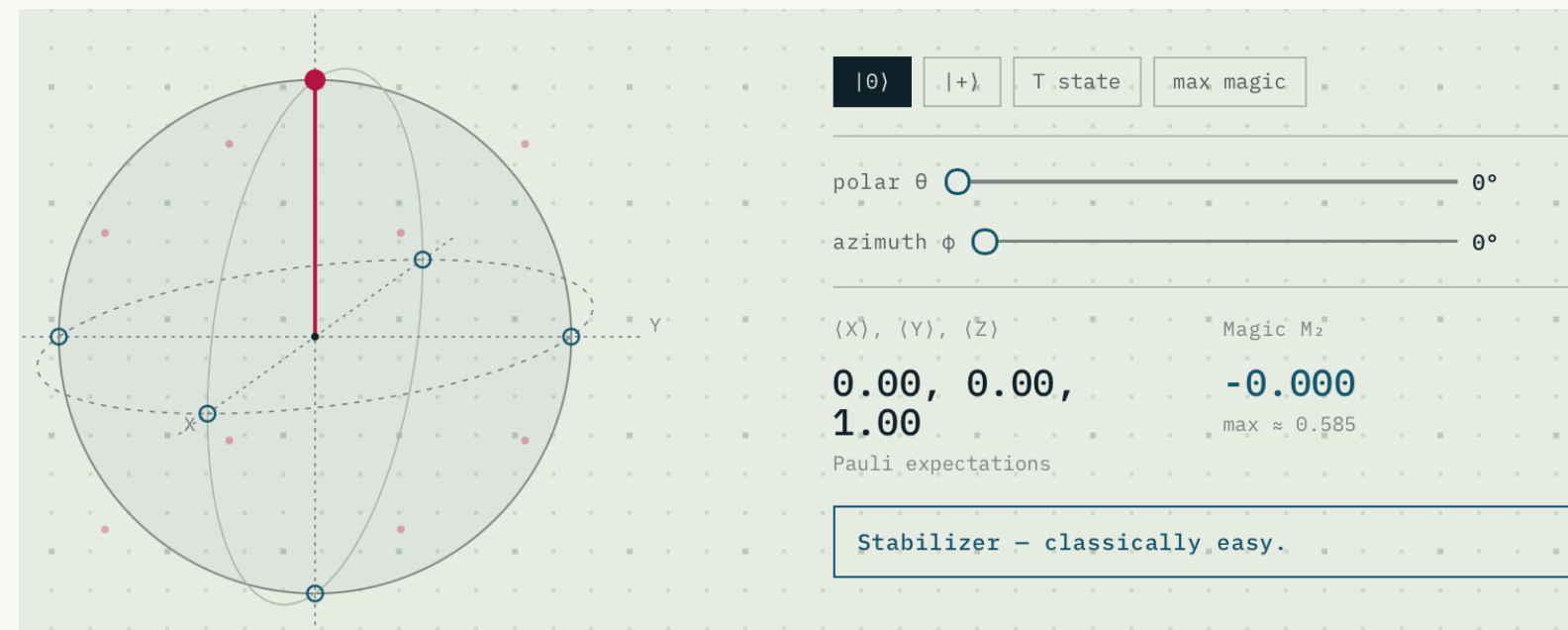
Sometimes we can still work classically! The real question becomes: *when can't we?*



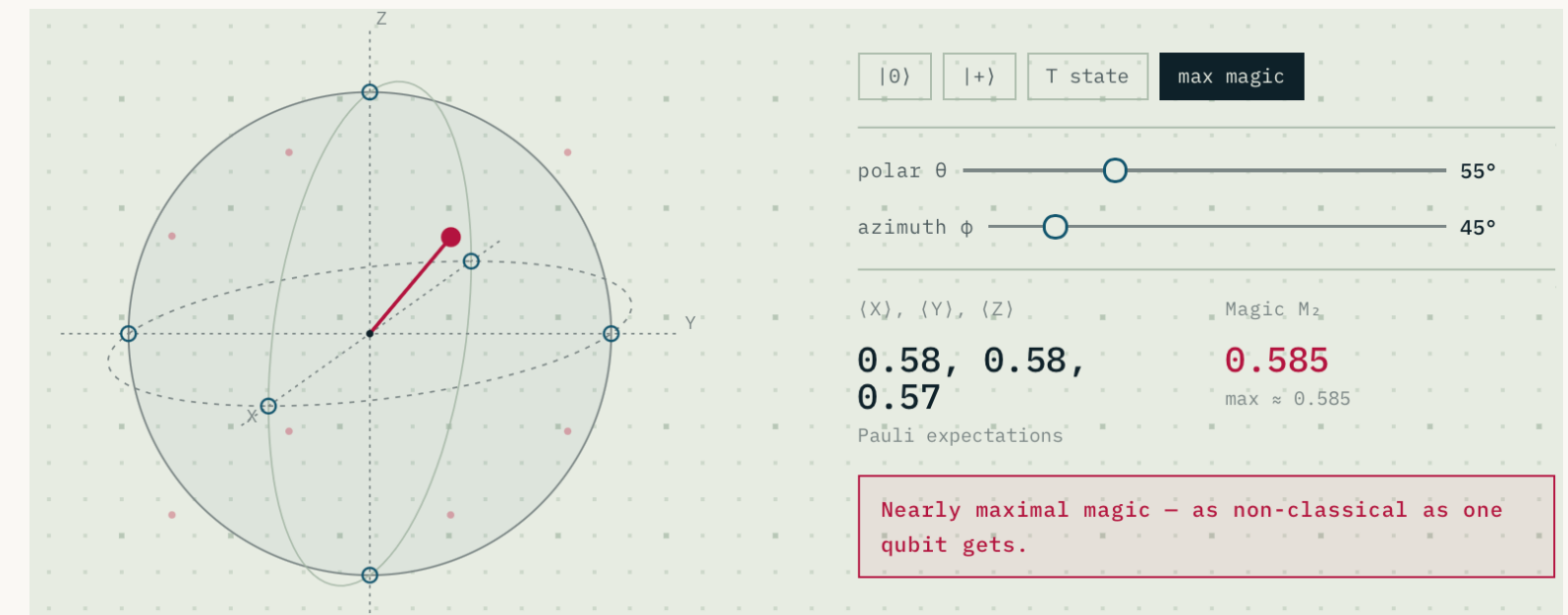
Relevant states occupy a vanishing corner of the full space.

2 · ONE QUBIT

Even for one qubit, some states are easy and some are hard



Easy — the pole $|0\rangle$



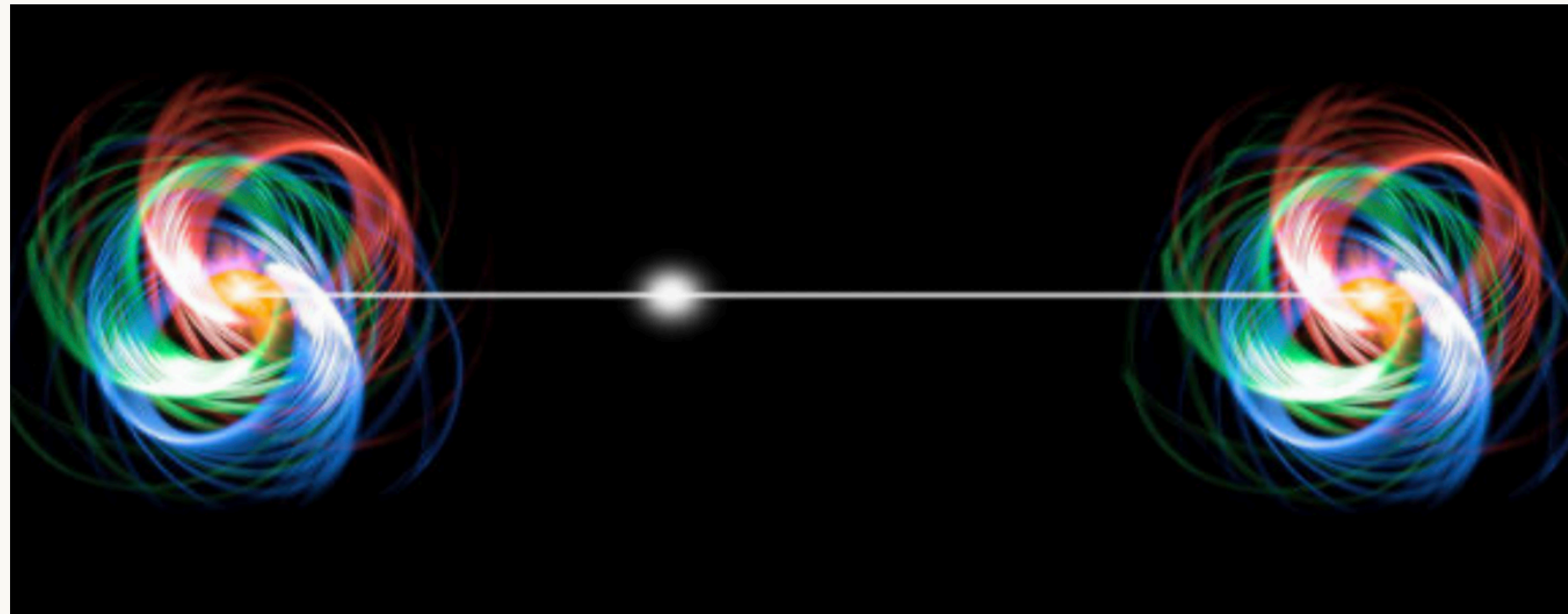
Hard — tilted off the special axes

The special directions ($|0\rangle, |+\rangle$) are **free**. Tilt *off* them and the state becomes hard — that cost is **magic**.

Natural guess - What can it be?

Entanglement

Reason: Must be some spooky action at a distance!



Entanglement is *not* what makes it hard

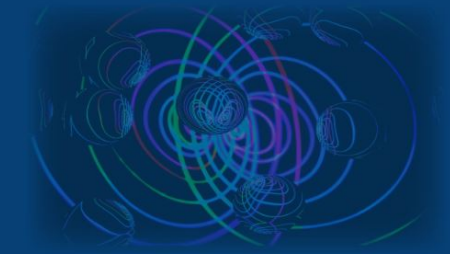
The natural guess is “more entanglement → harder.” The **Bell state** breaks it:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq |\psi_1\rangle \otimes |\psi_2\rangle \text{ — the } \textit{deviation} \text{ is a measure of entanglement}$$

- It is **maximally entangled** — the two qubits are as correlated as they can possibly be yet a classical computer can prepare it **efficiently**.

If maximal entanglement can still be easy, then some *other* ingredient must be what makes quantum computers powerful. That ingredient is magic!

2022 Nobel Prize
in Physics



Alain Aspect, John Clauser and Anton Zeilinger

Part II

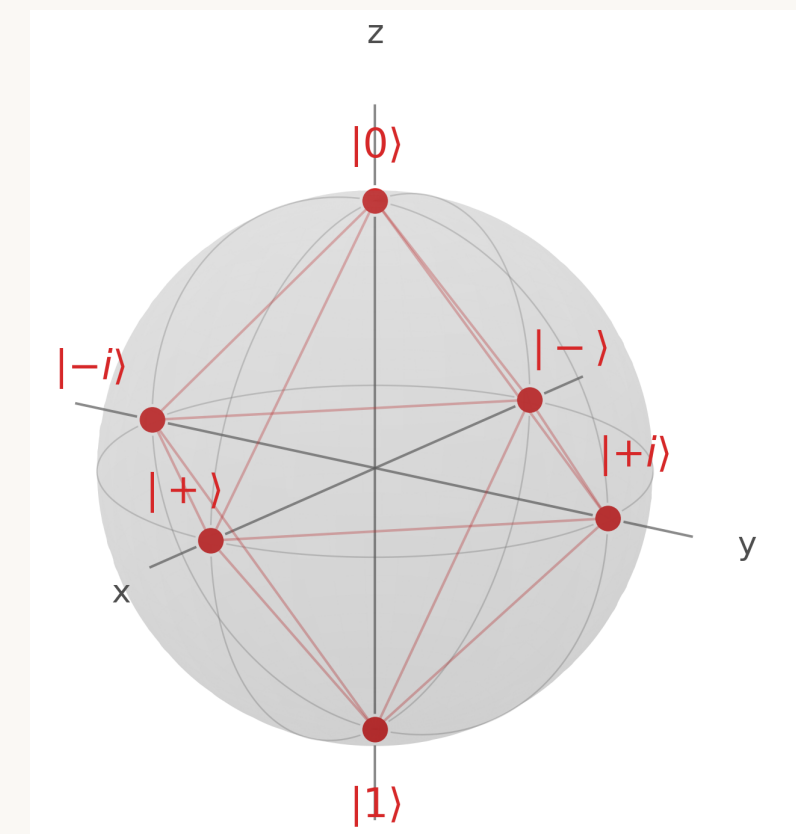
“Magic”: the missing ingredient

The resource that actually separates a quantum computer from a classical one
— and that we can measure.

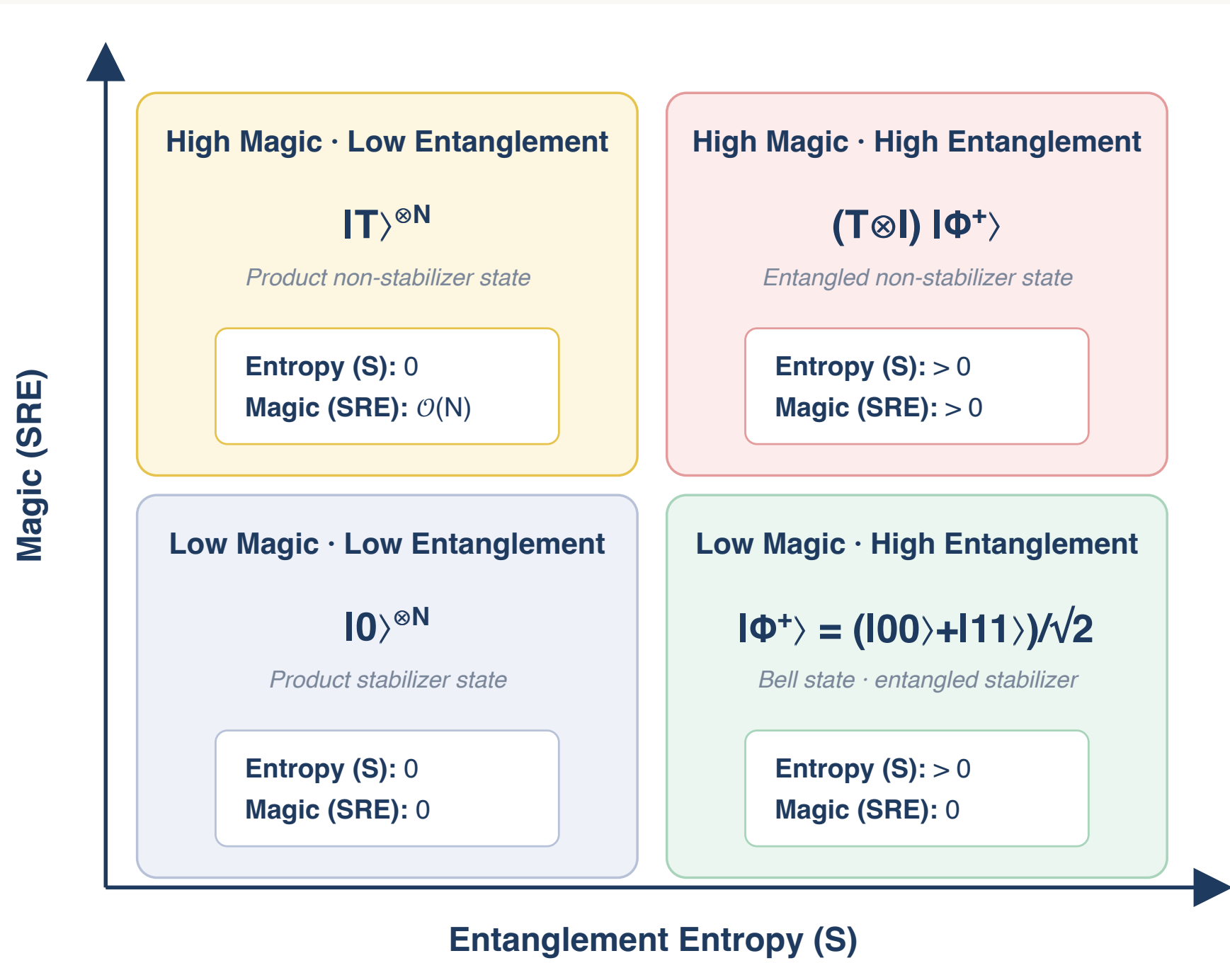
Magic is what is *not* on the vertices!

A special set of quantum gates — $\{H, S, CX\}$ — are exactly the ones a classical computer can keep up with.

- Circuits built only from these gates are **always classically simulable** (Gottesman–Knill theorem).
- Add **one** more gate — the **T-gate** — and we unlock **universal** quantum computing.
- Think of this as taking a plane to any city of the world, some are easy, some are harder to reach.



Entanglement and magic are independent resources



A state lives on a two-axis map: **entanglement** across, **magic** up.

- Bell state: max entanglement, **zero magic** → still easy.
- A truly **hard** state has **both**.

This is why we need a *new* ruler — entanglement does not capture the whole story.

Putting a number on magic

The modern answer (Leone *et al.*, 2021): the **Stabilizer Rényi Entropy** (SRE). Ask how “spread out” the state looks in the language of Pauli strings P (e.g. XYZ , $YZYZ$, $XYXYX$). For N qubits the set $\mathcal{P}_N$ has 4^N elements — for $N = 1, \{\mathbb{I}, X, Y, Z\}$. Most of them are $\log(1) = 0$

$$M_2 = -\log_2 \left(\frac{1}{2^N} \sum_{P \in \mathcal{P}_N} |\langle P \rangle|^4 \right) \quad ; \quad \langle P \rangle = \langle \psi | P | \psi \rangle$$

Zero for stabilizer states (the easy ones). **Positive** for T-states like $T^{\otimes N} |0\rangle^{\otimes N}$

CURSE OF DIMENSIONALITY

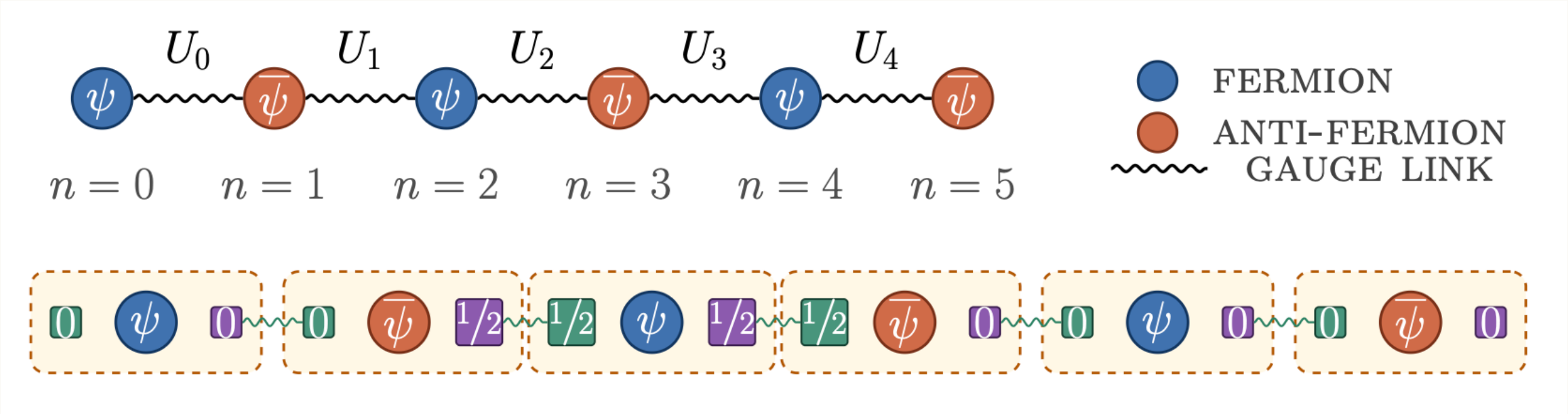
Computing it is exponentially hard — but it can be estimated by Monte Carlo sampling.

End Story

Strong force meets magic

Probing magic in the ground state of the strong force

A toy version of the strong force: 1+1 SU(2)



Sites carry matter (fermions/anti-fermions), links carry the gauge field; $j_{\max} = 1/2$.

Figure: arXiv:[2606.09971](#) — RGJ, Toga, Taher, Bakalov, Kemper

The simplest **non-Abelian** cousin of QCD — rich enough to be interesting.

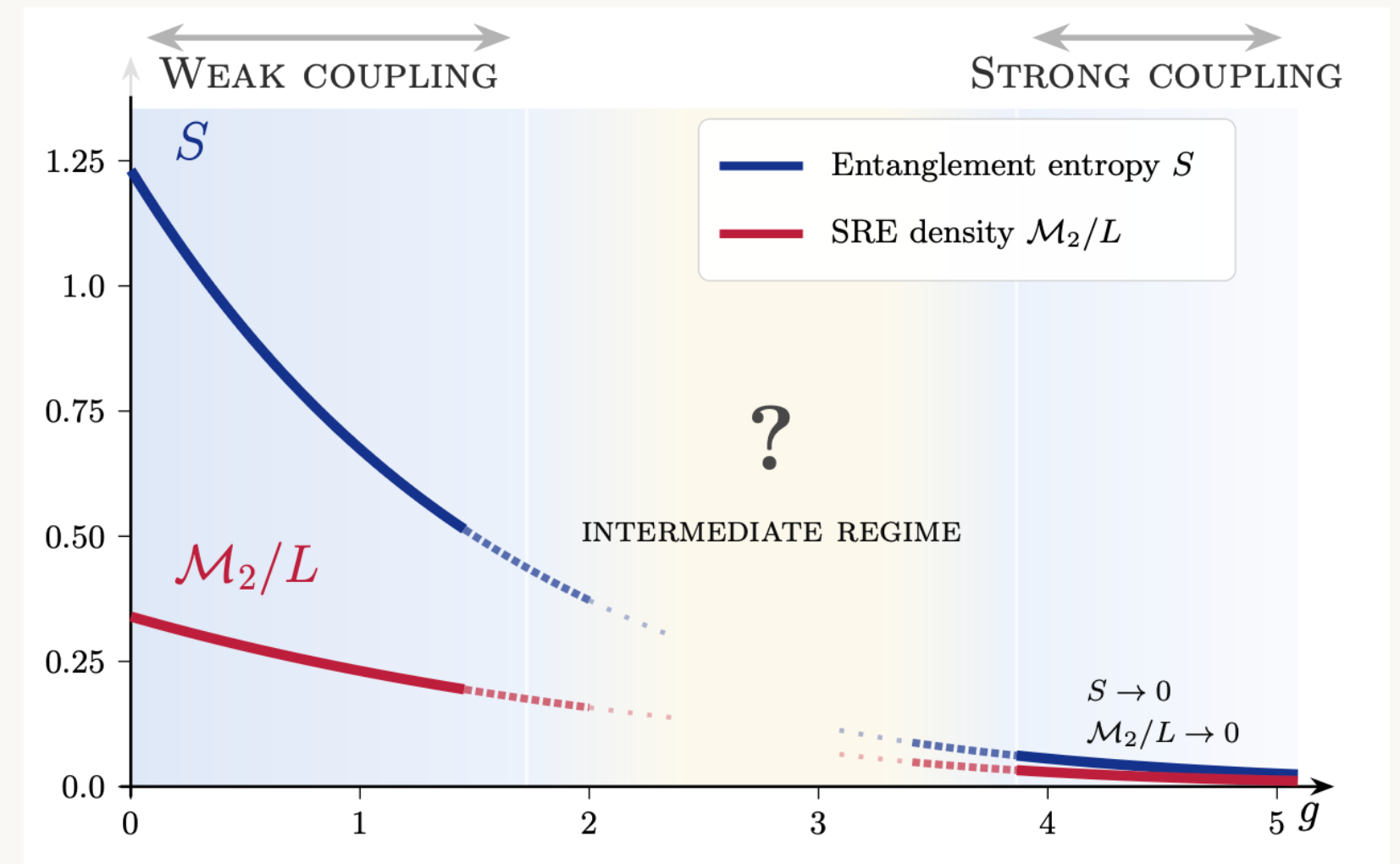
Only **6 of 36** local states are physical → a Hamiltonian on 6^L states, requiring **3L qubits**.

Before computing — what should we expect?

“... that one should never start a sentence without knowing its end.”

— Dirac to Bohr

- Small coupling $g \rightarrow 0$: near a **conformal** theory $\rightarrow$ **high magic** and **high entanglement**.
- Large coupling $g \rightarrow \infty$: a simple **product state** $\rightarrow$ both drop to zero.
- The interesting physics is in between.

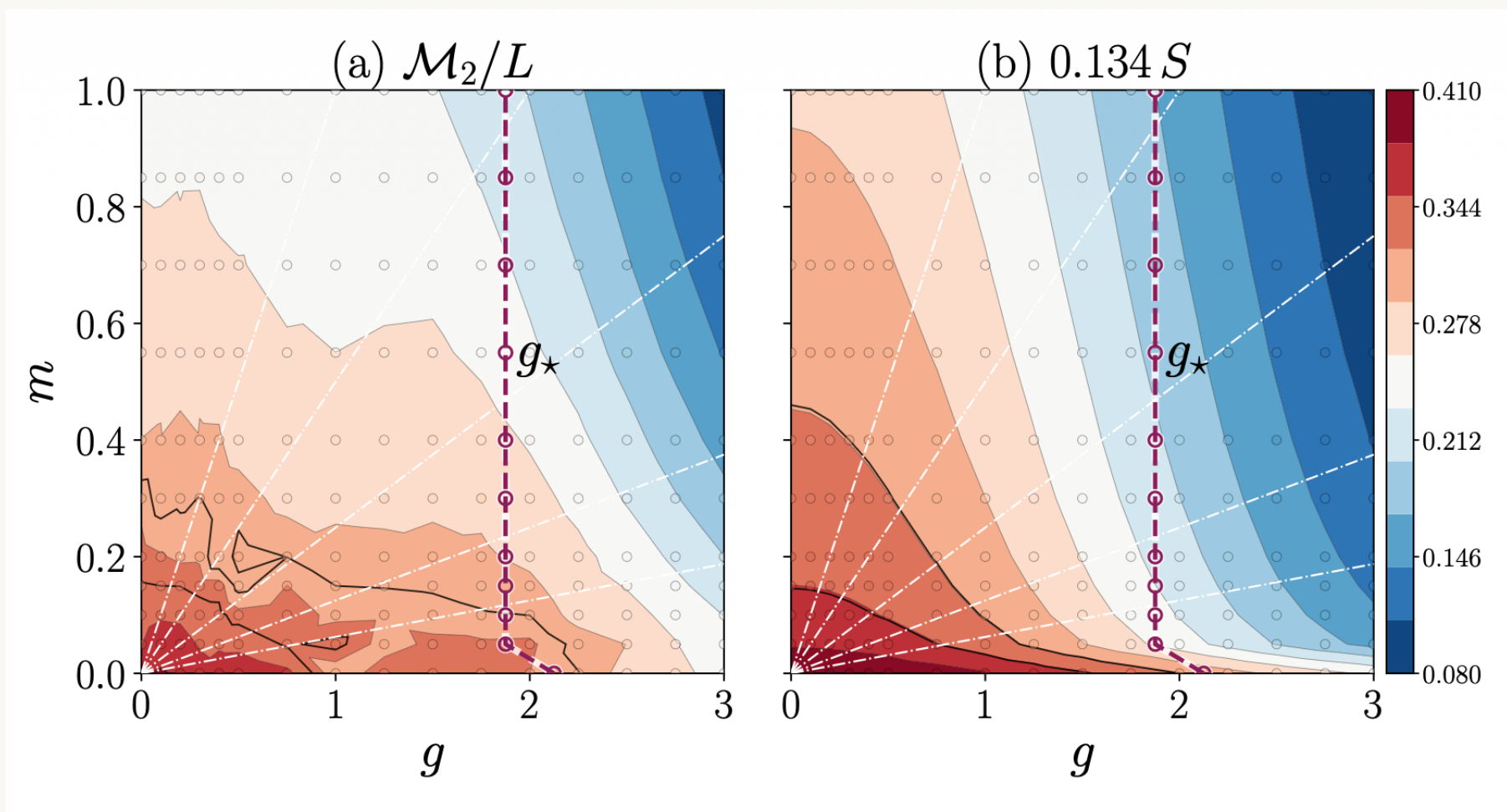


Cartoon expectation: entropy S and magic density fall as coupling grows.

Figure: arXiv:2606.09971 — RGJ, Toga, Taher, Bakalov, Kemper

6 · RESULTS

Magic and entanglement can go their separate ways



Mapped over the full phase diagram, the two heat maps **don't look the same**.

THE 1+1 BEHAVIOUR

Entanglement can be *consistently falling* while magic stays flat right up to the crossover. They are **genuinely different** quantum-resource measures — and the continuum limit $g \rightarrow 0$ is exactly where magic piles up. Quantum field theories are highly magical.

Magic density $m = \mathcal{M}_2/L$ (left) vs. entanglement S (right) across the mass–coupling plane.

Figure: arXiv:[2606.09971](#) — RGJ, Toga, Taher, Bakalov, Kemper

WHY CARE?

Quantum advantage road ahead!

- Reaching the **continuum limit** of these gauge theories means simulating states with **very high magic** — states that are genuinely non-classical. This is required to study the strong force on quantum computers!
- Magic gives us a **ruler for the quantum-advantage frontier** : which physics truly needs a quantum computer, and which doesn't. Note the only ruler!

IN ONE SENTENCE

Magic turns a vague hope — “quantum computers will help with the strong force” — into a *measurable* statement about how hard the job really is. Simulating real-time dynamics of QCD is most likely a very hard problem for quantum computers.

SUMMARY

Two take-away messages

- 1 Quantum hardness $\neq$ entanglement. The relevant resource is **magic** — and we can measure it. We want to push this direction for lattice gauge theories, up to 3+1-dimensional QCD.
- 2 The **continuum limit** of gauge theories carries high magic — simulating the strong force is genuinely quantum-hard.

Thank you

Questions?

Based on **arXiv:2606.09971** + **arXiv:2609.XXXXX**

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